PRACTICE SHEET

1. What is the interval in which the function $f(x) = \sqrt{9 - x^2}$ is increasing? ($f(x) > 0$)
(a) $0 < x < 3$ (b) $-3 < x < 0$
(c) $0 < x < 9$ (d) $-3 < x < 3$

2. A wire 34 cm long is to be bent in the form of a quadrilateral of which each angle is 90°. What is the maximum area which can be enclosed inside the quadrilateral?
(a) $68cm^2$ (b) $70cm^2$
(c) 71.25 cm^2 (d) 72.25 cm^2

3. Which one of the following is correct? The function $f(x) = (x - 1)e^x + 1$ is
(a) Negative for all $x > 0$
(b) Positive for all $x > 0$
(c) Increasing for all x
(d) Decreasing for all x

4. The motion of a particle is described as $s = 2 - 3t + 4t^3$. What is the acceleration of the particle at the point where its velocity is zero?
(a) 0 (b) 4
(c) 8 unit (d) 12 unit

5. What is the product of two parts of 20, such that the product of one part and the cube of the other is maximum?
(a) 75 (b) 91

terms of the side x is given by which one of the following function?
(a) $f(x) = (40 / x) + 2x^2$
(b) $f(x) = (40 / x) + x^2$
(c) $f(x) = (40 / x) + x^2$
(d) $f(x) = (60 / x) + 2x$

13. $f(x) = \cos x$ is monotonic decreasing under which one of the following conditions?
(a) $0 < x < \frac{\pi}{2}$ only (b) $\frac{\pi}{2} < x < \pi$ only
(c) $0 < x < \pi$ (d) $0 < x < 2\pi$

14. What is the minimum value of $2x^2 - 3x + 5$?
(a) 0 (b) 3/4
(c) 31/4 (d) 31/8

15. **Assertion (A):** The tangent to the curve $y = x^3 - x^2 - x + 2$ at (1, 1) is parallel to the x – axis.
Reason (R): The slope of the tangent to the curve at (1, 1) is zero
(a)Both A and R are true and R is the correct explanation of A
(b)Both A and R are true but R is not the correct explanation of A
(c)A is true but R is false
(d)A is false but R is false